

Analysis:

Chapter 1: The real number system

②. Real numbers and their Properties:

- Addition
- Multiplication
- $\mathbb{R}$ is totally ordered by the order relation $\leq$; it verifies the assumptions:

1. Reflexive: $\forall x \in \mathbb{R} : x \leq x$.

2. Antisymmetric:

$$\forall x, y \in \mathbb{R} : x \leq y \wedge y \leq x \Rightarrow x = y.$$

3. The transitive law:

$$\text{if } x \leq y \wedge y \leq z \Rightarrow x \leq z.$$

④. Absolute Value:

* properties:

- $\forall x \in \mathbb{R} : |x| \geq 0$.
- $\forall x \in \mathbb{R} : |-x| = |x|$.
- $\forall x \in \mathbb{R} : |x| = 0 \Rightarrow x = 0$.
- $\forall x \in \mathbb{R} ; \forall a \in \mathbb{R}_+,$
 $|x| \leq a \Leftrightarrow -a \leq x \leq a$.
- $\forall x \in \mathbb{R}, \forall a \in \mathbb{R}_+,$
 $|x| > a \Leftrightarrow x > a \vee x < -a$.
- $\forall x \in \mathbb{R}, \forall y \in \mathbb{R}^* : \left| \frac{x}{y} \right| = \frac{|x|}{|y|}$.
- $\forall x, y \in \mathbb{R} : |x+y| \leq |x| + |y|$.
(triangle inequality)
- $\forall x, y \in \mathbb{R} : |xy| = |x| \cdot |y|$.
- $\forall x, y \in \mathbb{R} : ||x| - |y|| \leq |x - y|$.
- $\forall x \in \mathbb{R} : \sqrt{x^2} = |x| \wedge |x|^2 = x^2$.
- $\forall x \in \mathbb{R} : -|x| \leq x \leq |x|$.

⑤. Bounded Sets:

• Upper and lower bounds:

$$\Leftrightarrow \left(\begin{array}{l} M \text{ is an} \\ \text{upper} \\ \text{bound of } A \end{array} \right) \Leftrightarrow (\forall a \in A : a \leq M)$$

$$\Leftrightarrow \left(\begin{array}{l} A \text{ is bounded} \\ \text{above} \end{array} \right) \Leftrightarrow (A \text{ has an upper bound})$$

$$\Leftrightarrow \left(\begin{array}{l} m \text{ is a lower} \\ \text{bound of } A \end{array} \right) \Leftrightarrow (\forall a \in A : m \leq a)$$

$$\Leftrightarrow \left(\begin{array}{l} A \text{ is bounded} \\ \text{below} \end{array} \right) \Leftrightarrow (A \text{ has a lower bound})$$

$$\Leftrightarrow (A \text{ is bounded}) \Leftrightarrow (A \text{ is bounded above and below})$$

$$\Leftrightarrow (A \text{ is unbounded}) \Leftrightarrow (A \text{ fails to be bounded})$$

$$\Leftrightarrow \left(\begin{array}{l} A \text{ is} \\ \text{unbounded above} \\ \text{or} \\ \text{unbounded below} \end{array} \right) \Leftrightarrow \left(\begin{array}{l} A \text{ fails to be} \\ \text{bounded above,} \\ \text{bounded below} \end{array} \right)$$

• Maximum and Minimum:

$$\Leftrightarrow \left(\begin{array}{l} M \text{ is an upper} \\ \text{bound} \\ M \in A \end{array} \right) \Rightarrow (\text{Max}(A) = M)$$

$$\Leftrightarrow \left(\begin{array}{l} m \text{ is a lower} \\ \text{bound} \\ m \in A \end{array} \right) \Rightarrow (\text{min}(A) = m)$$

• Supremum and Infimum:

$$\Leftrightarrow \left(\begin{array}{l} \lambda \text{ is an upper} \\ \text{bound of } A \\ \lambda \leq m \\ \text{(the least} \\ \text{upper bound)} \end{array} \right) \Rightarrow (\text{Sup}(A) = \lambda)$$

~~Prop.~~ $\text{Sup}(A)$ is unique.

- $\text{Sup}(A) = \lambda$, whether $\lambda \in A$ or $\lambda \notin A$.
- if $\text{Sup}(A) \in A$, $\text{Sup}(A) = \max(A)$.
- $\forall x \in A : \text{Sup}(A) \geq x$.
- $\forall \varepsilon > 0, \exists x_\varepsilon \in A :$

$$\text{Sup}(A) - \varepsilon < x_\varepsilon \leq \text{Sup}(A)$$

⑤. $\left(\begin{array}{l} \lambda \text{ is a lower} \\ \text{bound of } A \\ \lambda \geq m \\ \text{(the greatest} \\ \text{lower)} \end{array} \right) \Rightarrow (\inf(A) = \lambda)$

→ $\inf(A)$ is unique.

- $\inf(A) = \lambda$, whether $\lambda \in A$ or $\lambda \notin A$.
- if $\inf(A) \in A$, $\inf(A) = \min(A)$.
- $\forall x \in A : \inf(A) \leq x$.
- $\forall \varepsilon > 0, \exists x_\varepsilon \in A : \inf(A) \leq x_\varepsilon < \inf(A) + \varepsilon$

⑥. The Archimedean property:

- $\forall a, b \in \mathbb{R}_+, \exists n \in \mathbb{N}: na > b$
- * $\forall x \in \mathbb{R}, \exists n \in \mathbb{N}: n > x$

⑦. The integer part:

- $\forall n \in \mathbb{R} : E(n) = [n] = \max \{n \in \mathbb{Z} : n \leq n\}$
- * $\forall n \in \mathbb{R} : n = E(n) + r \quad (0 \leq r < 1)$
(r is the decimal part of n)

* properties:

- $\forall n \in \mathbb{R} : E(n) \in \mathbb{Z}$.
- if $n \in \mathbb{Z} : E(n) = n$
- $\forall n \in \mathbb{R} : E(n) \leq n < E(n) + 1$
- $\forall n, y \in \mathbb{R} :$

$$E(n) + E(y) \leq E(n+y) < E(n) + E(y) + 1$$

(8). Density of the rational numbers
(RATIS)

- $\forall n, y \in \mathbb{R} : n < y \Rightarrow \exists r \in \mathbb{Q} : n < r < y$
($\mathbb{Q}$ is dense in $\mathbb{R}$)

②. Intervals and neighborhoods:

- Finite intervals:

- $\forall a, b \in \mathbb{R}, a < b :]a, b[= \{n, a < n < b\}$
- $\forall a, b \in \mathbb{R}, a < b : [a, b[= \{n, a \leq n < b\}$
- $\forall a, b \in \mathbb{R}, a < b :$
 - $]a, b] = \{n, a < n \leq b\}$
 - $[a, b] = \{n, a \leq n \leq b\}$

- all the intervals above are called finite intervals, a, b are called end points (a : left hand end point
 b : right " " " ")
any point in the interval $\neq a, b$ is called an interior point of the interval, $\frac{1}{2}(a+b)$ is the mid point.

- infinite intervals:

- $]a; +\infty[: a < n;$
 - $[a; +\infty[: a \leq n;$
 - $] -\infty; a[: n < a;$
 - $] -\infty, a] : n \leq a;$
 - $] -\infty; +\infty[: n \in \mathbb{R};$
- (a is an end point)

- a neighborhood of a point $(\in \mathbb{R})$ is an open interval of the form:

$$]a - \epsilon, a + \epsilon[, \epsilon \in \mathbb{R}_+$$

- $\exists a - \varepsilon, a + \varepsilon [: |n - a| < \varepsilon$
 $(a \text{ is the mid point of each of its neighbohoods})$

$x-1 < E(x) \leq x$

 it isn't
 considered
 as property
 (it's found in)

$$E(n) + E(y) \leq E(n+y) < E(n) + E(y) + 1$$

Chapter 2: Complex Numbers (z → complex number)

① Definitions:

• complex number: $\frac{a}{1} + \frac{bi}{i}$
 the real part $\frac{a}{1}$ the imaginary part $\frac{bi}{i}$

• $\begin{cases} a \neq 0 \\ b = 0 \end{cases} \Rightarrow z = a$ (a real number)

• $\begin{cases} a = 0 \\ b \neq 0 \end{cases} \Rightarrow z = bi$ (purely imaginary number)

• $\begin{cases} a = 0 \\ b = 0 \end{cases} \Rightarrow z = 0$

② Algebraic operations on complex numbers:

→ Addition:

$$z_1 + z_2 = (a+c) + (b+d)i$$

→ Subtraction:

$$z_1 - z_2 = (a-c) + (b-d)i$$

→ Multiplication:

$$z_1 \cdot z_2 = (ac + bd) + (ad + bc)i$$

→ Division:

$$\frac{z_1}{z_2} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2}i$$

→ equality:

$$z_1 = z_2 \Leftrightarrow \begin{cases} a = c \\ b = d \end{cases}$$

③ Conjugate of a complex number:

* $\bar{z} = a - bi$ is the conjugate of z .

• if $z \in \mathbb{R} \Rightarrow z = \bar{z}$

• if $z = bi$ (purely imaginary number)
 $\Rightarrow \bar{z} = -z$

$$\bar{\bar{z}} = z$$

$$\forall z, w \in \mathbb{C} : \overline{z+w} = \bar{z} + \bar{w}$$

$$\forall z, w \in \mathbb{C} : \overline{z \cdot w} = \bar{z} \cdot \bar{w}$$

$$\forall z \in \mathbb{C} : z + \bar{z} = 2 \cdot \text{Re } z$$

the real part of z

$$\forall z \in \mathbb{C} : z - \bar{z} = 2i \cdot \text{Im } z$$

④ Modulus of a complex number:

* $|z| = \sqrt{a^2 + b^2}$ is the modulus of z .

$$|z| = 0 \Rightarrow z = 0$$

$$|z| = |\bar{z}| \quad \wedge \quad |z| = |-z|$$

$$\text{if } \text{Im } z = 0 \Rightarrow z^2 = |z|^2$$

$$\forall z, w \in \mathbb{C} : |zw| = |z| \cdot |w|$$

$$\left| \frac{z}{w} \right| = \frac{|z|}{|w|} \quad (w \neq 0)$$

⑤ Geometrical representation of complex numbers:

• a point.

• z can be considered as a couple ordered of real numbers.

$$z = a + bi \rightsquigarrow M(a, b)$$

• $(Ox) \rightarrow$ real axis.

• $(Oy) \rightarrow$ imaginary axis.

* if M_1 is the point of affix z , the point M_2 of affix $\bar{z}$ is the symmetric of M_1 with respect to x -axis.

• a circle:

• circle radius r , center $z_0 = x_0 + iy_0$ defined by the equation:

$$|z - z_0| = r$$

• Segment:

• the line segment connecting two complex points z_0 and z_1 is the set defined by:

$$\{z \in \mathbb{C}, z = (1-t)z_0 + tz_1 \text{ such that } t \in [0, 1]\}$$

• Graph: $z \in [a, b]$, f is continuous function.

the graph of f is defined as the set:

$$G_f = \{z \in \mathbb{C}; z = n + i f(n) \text{ with } n \in [a, b]\}$$

⑥ - Polar form:

$$P(a, b): z = a + bi \quad (a, b \in \mathbb{R})$$

OP makes θ with the positive direction of x-axis, $PM \perp OX$, $OP = r$, then:

$$a = r \cos \theta, \quad b = r \sin \theta$$

• $z = r(\cos \theta + i \sin \theta)$ ← the polar form

$$* r = \sqrt{a^2 + b^2}$$

$$* \tan \theta = \frac{b}{a}$$

• if $-\pi \leq \theta \leq \pi$, θ is called the principal argument: $\arg z$,

$$\theta = \arg z + 2k\pi, \quad k \in \mathbb{Z}$$

$$• \arg(z z') = \arg(z) + \arg(z') + 2k\pi$$

$$• \arg\left(\frac{1}{z}\right) = \arg(\bar{z}) = -\arg(z) + 2k\pi$$

$$• \arg\left(\frac{z}{z'}\right) = \arg(z) - \arg(z') + 2k\pi$$

• $z = r(\cos \theta + i \sin \theta) = re^{i\theta}$ ← the exponential form

⑦ - De Moivre formula:

$$• z^n = (r(\cos \theta + i \sin \theta))^n = r^n (\cos(n\theta) + i \sin(n\theta))$$

$$\text{if } z^n = a + ib \Rightarrow z = (a + ib)^{\frac{1}{n}}$$

using De Moivre formula, we get:

$$z = (r(\cos \theta + i \sin \theta))^{\frac{1}{n}}$$

$$= r^{\frac{1}{n}} \left(\cos\left(\frac{\theta + 2k\pi}{n}\right) + i \sin\left(\frac{\theta + 2k\pi}{n}\right) \right)$$

$$k = 0, 1, \dots, n-1$$

and we conclude the existence of n -th roots difference of $a + ib$.

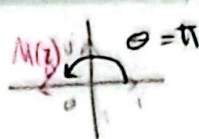
old Remarks:

$$• i^2 = -1 \quad • -i^2 = 1$$

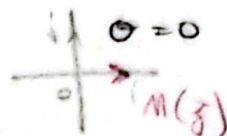
$$• (1+i)^2 = 2i \quad • (1-i)^2 = -2i$$

• Les familles:

→ 1^{ère} famille:

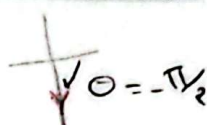


$$z = -\alpha$$

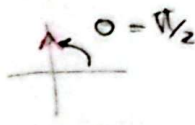


$$z = \alpha$$

→ 2^{ème} famille: $\{ |z| = |\alpha| \}$

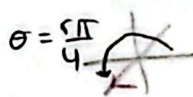


$$z = -\alpha i$$

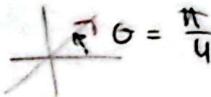


$$z = \alpha i$$

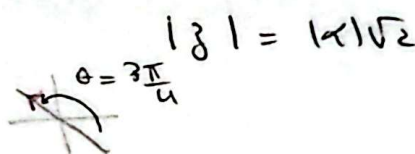
→ 3^{ème} famille:



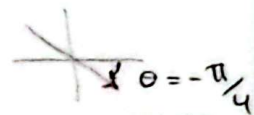
$$z = -\alpha - \alpha i$$



$$z = \alpha + \alpha i$$



$$z = -\alpha + \alpha i$$



$$z = \alpha - \alpha i$$

→ 4^{ème} famille:

$$z = \mp \alpha + \mp \alpha \sqrt{3} i$$

la famille de $\pi/3$ ($\pi/3$ / $2\pi/3$ / $4\pi/3$ / $-pi/3$)

$$\frac{\operatorname{Im}(z)}{\operatorname{Re}(z)} = \mp \sqrt{3}$$

$$(|\operatorname{Im}(z)| > |\operatorname{Re}(z)|)$$

→ 5^{ème} famille:

$$z = \mp \alpha \sqrt{3} \mp \alpha i$$

la famille de $\pi/6$ ($\pi/6$ / $5\pi/6$ / $7\pi/6$ / $-pi/6$)

$$\frac{\operatorname{Re}(z)}{\operatorname{Im}(z)} = \mp \sqrt{3}$$

$$(|\operatorname{Re}(z)| > |\operatorname{Im}(z)|)$$

Chapter 3: Sequences

① Bounded sequences:

- $2_n \leq M \Rightarrow$ M is an upper bound of (2_n)
(2_n) is bounded above
- $m \leq 2_n \Rightarrow$ m is a lower bound
(2_n) is bounded below
- 2_n is bounded $\Rightarrow$ bounded above + below
- (2_n is bounded) $\Leftrightarrow (\exists k > 0, |2_n| \leq k, \forall n \in \mathbb{N})$
- increasing $\Rightarrow 2_n \leq 2_{n+1}$
(strict $<$)
- decreasing $\Rightarrow 2_n \geq 2_{n+1}$
(strict $>$)
- constant $\Rightarrow$ increa + decrea

② The convergent sequences:

- $\forall n \in \mathbb{N}, n \geq N(\epsilon) \Rightarrow |2_n - l| < \epsilon$
converge $\neq$ diverge
- (2_n) is divergent $\Rightarrow$ $\begin{cases} \lim_{n \rightarrow \infty} 2_n = +\infty \\ \text{or} \\ 2_n \text{ has two different limits} \end{cases}$
- (2_n) converges $\Rightarrow$ (2_n) is bounded
one side $\rightarrow$ counter example.
- Arithmetic operations: $(-1)^n$
 $\lim_{n \rightarrow \infty} 2_n = k, \lim_{n \rightarrow \infty} v_n = L$
- $w_n = 2_n + v_n \Rightarrow \lim_{n \rightarrow \infty} w_n = k + L$
(multiplication + division)

③ Limit theorems:

- $\lim_{n \rightarrow \infty} 2_n = 0$
+ v_n is bounded $\Rightarrow \lim_{n \rightarrow \infty} 2_n v_n = 0$
- $\lim_{n \rightarrow \infty} 2_n = k / \lim_{n \rightarrow \infty} v_n = L$
 $\exists n_0 \in \mathbb{N}: 2_n \leq v_n, \forall n \geq n_0$
 $\Rightarrow k \leq L$

$$2_n \leq w_n \leq v_n \text{ and } l \text{ is the limit of } v_n \text{ and } 2_n \text{ converge to } l$$

- $\lim_{n \rightarrow \infty} 2_n = k : \begin{cases} 2_n \geq 0 \Rightarrow k \geq 0 \\ 2_n \leq 0 \Rightarrow k \leq 0 \end{cases}$
- $2_n \leq v_n : \begin{cases} 2_n \rightarrow +\infty \Rightarrow v_n \rightarrow +\infty \\ v_n \rightarrow -\infty \Rightarrow 2_n \rightarrow -\infty \end{cases}$

④ Monotonic sequences:

- $2_0 = \inf \{2_n, n \in \mathbb{N}\}$
- $\lim_{n \rightarrow \infty} 2_n = \sup \{2_n, n \in \mathbb{N}\}$
if increasing and bounded above
(the contrary $\rightarrow$ the contrary)
- (2_n) is increasing, not bounded above
or the contrary $\Rightarrow$ (2_n) is divergent

chapter 4: functions.

① - Basic Notions: $f: A \rightarrow B$

• the range of f is defined as:

$$H(f) = \{y = f(n) : n \in D(f)\}$$

• the set: $G_f = \{(n, f(n)) : n \in D\}$

is called the graph of f . ($G_f \subset \mathbb{R}^2$)

• $f = g$, if: $\begin{cases} D_f = D_g \\ \forall n \in D_f : f(n) = g(n) \end{cases}$

• f is bounded below on $D_f \Rightarrow \exists m \in \mathbb{R}, f(n) \geq m, \forall n \in D_f$.

• f is bounded above on $D_f \Rightarrow \exists M \in \mathbb{R}, f(n) \leq M, \forall n \in D_f$.

• f is bounded $\Rightarrow f$ is bounded below and above.

1) $f(n_1) \leq f(n_2) \Rightarrow f$ is increasing on A

2) $f(n_1) < f(n_2) \Rightarrow f$ is strictly increasing on A .

3) $f(n_1) \geq f(n_2) \Rightarrow f$ is decreasing on A

4) $f(n_1) > f(n_2) \Rightarrow f$ is strictly decreasing on A .

1 $\rightarrow$ 4 $\Rightarrow$ monotone
2 or 4 $\Rightarrow$ strictly monotone

• proving the monotonicity of f :

$\rightarrow f$ is increasing if $\frac{f(n) - f(y)}{n - y} > 0$
 $\forall n, y \in A, n \neq y$

$\rightarrow f$ is decreasing if $\frac{f(n) - f(y)}{n - y} < 0$
 $\forall n, y \in A, n \neq y$.

$\rightarrow$ Composition of functions:

$$f: A \rightarrow B, g: B \rightarrow C, g \circ f: A \rightarrow C$$

$$\bullet \forall n \in A : (g \circ f)(n) = g(f(n))$$

$$\bullet (h \circ g) \circ f = h \circ (g \circ f)$$

$$\bullet \begin{matrix} f \nearrow, g \nearrow \\ \text{or} \\ f \downarrow, g \downarrow \end{matrix} \Rightarrow g \circ f \nearrow$$

$$\bullet f \nearrow, g \downarrow$$

$$f \downarrow, g \nearrow \Rightarrow g \circ f \downarrow$$

$\rightarrow$ Special classes of functions:

* even and odd functions:

$$f: D_f \rightarrow \mathbb{R}$$

• f is even $\Rightarrow \forall n \in D_f : \begin{cases} -n \in D_f \\ f(-n) = f(n) \end{cases}$

• f is odd $\Rightarrow \forall n \in D_f : \begin{cases} -n \in D_f \\ f(-n) = -f(n) \end{cases}$

• Graph of an even function is symmetric with respect to y-axis.

• Graph of an odd function is symmetric with respect to the origin.

• the domain of the two is always symmetric with respect to the origin.

* periodic functions:

• f is periodic $\Rightarrow \exists p \in \mathbb{R}^* : \begin{cases} x \in D_f \Rightarrow x+p \in D_f \\ \forall n \in D_f : f(n+p) = f(n) \end{cases}$

• p is called a period of f .

• the smallest positive period is called primitive.

• if f is periodic with p , $g(f(n))$ is periodic with the same p .

• if f is periodic with p and $a \in \mathbb{R}^*$ the function $g(n) = f(an)$ is periodic with $\frac{p}{a}$.

②. Limit of function:

$$1) \lim_{n \rightarrow x_0} f(n) = l$$

the limit l of f exists at $x = x_0$ if:

$$\forall \epsilon > 0, \exists \delta > 0, \forall n \in D_f :$$

$$|n - x_0| < \delta \Rightarrow |f(n) - l| < \epsilon$$

$$2) \lim_{n \rightarrow x_0} f(n) = +\infty :$$

$$\forall A > 0, \exists \delta > 0, \forall n \in D_f :$$

$$|n - x_0| < \delta \Rightarrow f(n) > A$$

$$3) \lim_{n \rightarrow x_0} f(n) = -\infty :$$

$$\forall B < 0, \exists \delta > 0, \forall n \in D_f :$$

$$|n - x_0| < \delta \Rightarrow f(n) < B$$

$$4) \lim_{n \rightarrow +\infty} f(n) = l :$$

$$\forall \varepsilon > 0, \exists A > 0, \forall n \in D_f :$$

$$n > A \Rightarrow |f(n) - l| < \varepsilon$$

$$5) \lim_{n \rightarrow -\infty} f(n) = l :$$

$$\forall \varepsilon > 0, \exists B < 0, \forall n \in D_f :$$

$$n < B \Rightarrow |f(n) - l| < \varepsilon$$

$$6) \lim_{n \rightarrow +\infty} f(n) = +\infty :$$

$$\forall A > 0, \exists B > 0, \forall n \in D_f :$$

$$n > B \Rightarrow f(n) > A$$

$$7) \lim_{n \rightarrow -\infty} f(n) = -\infty :$$

$$\forall A < 0, \exists B < 0, \forall n \in D_f :$$

$$n < B \Rightarrow f(n) < A$$

$$8) \lim_{n \rightarrow +\infty} f(n) = -\infty :$$

$$\forall A < 0, \exists B > 0, \forall n \in D_f :$$

$$n > B \Rightarrow f(n) < A$$

$$9) \lim_{n \rightarrow -\infty} f(n) = +\infty :$$

$$\forall A > 0, \exists B < 0, \forall n \in D_f :$$

$$n < B \Rightarrow f(n) > A$$

$$10) \lim_{n \rightarrow x_0} f(n) = l : \quad (\text{limit from the right})$$

$$\forall \varepsilon > 0, \exists \delta > 0, \forall n \in D_f :$$

$$0 < n - x_0 < \delta \Rightarrow |f(n) - l| < \varepsilon$$

$$11) \lim_{n \rightarrow x_0} f(n) = l : \quad (\text{limit from the left})$$

$$\forall \varepsilon > 0, \exists \delta > 0, \forall n \in D_f :$$

$$|n - x_0| < \delta \text{ and } n < x_0 \Rightarrow |f(n) - l| < \varepsilon$$

$$\left(\lim_{n \rightarrow x_0} f(n) = l \right) \Leftrightarrow \left(\lim_{n \nearrow x_0} f(n) = \lim_{n \searrow x_0} f(n) = l \right)$$

f, g, h are functions defined in a neighborhood of $x_0 \in \mathbb{R}$;

$$\begin{cases} f(n) \geq 0 \\ \lim_{n \rightarrow x_0} f(n) = l \end{cases} \Rightarrow l \geq 0$$

$$\begin{cases} f(n) \geq g(n) \\ \lim_{n \rightarrow x_0} f(n) = l \\ \lim_{n \rightarrow x_0} g(n) = l' \end{cases} \Rightarrow l \geq l'$$

$$\begin{cases} g(n) \leq f(n) \leq h(n) \\ \lim_{n \rightarrow x_0} g(n) = \lim_{n \rightarrow x_0} h(n) = l \end{cases} \Rightarrow \lim_{n \rightarrow x_0} f(n) = l$$

f, g two functions defined in a neighborhood of $x_0 \in \mathbb{R}$

$$\lim_{n \rightarrow x_0} f(n) = 0 \quad \text{ex. } \lim_{n \rightarrow 0} \sin(n)$$

$$g \text{ is bounded } N(x_0) \quad \text{ex. } \lim_{n \rightarrow 0} \sin(n) \text{ is bounded}$$

$$\Rightarrow \lim_{n \rightarrow x_0} f(n) g(n) = 0 \quad \text{ex. } \lim_{n \rightarrow 0} \sin(n) \leq 1$$

f defined in a neighborhood of $x_0 \in \mathbb{R}$
 g " " " of $l \in \mathbb{R}$

$$\lim_{n \rightarrow x_0} f(n) = l \text{ and } \lim_{n \rightarrow l} g(n) = m$$

$$\Rightarrow \lim_{n \rightarrow x_0} (g \circ f)(n) = m$$

→ limit of function by sequential approach

$$x_0 \in \mathbb{R} \cup \{-\infty, +\infty\}, l \in \mathbb{R} \cup \{-\infty, +\infty\}$$

$$\lim_{n \rightarrow x_0} f(n) = l \quad (1)$$

$$\text{For all } (x_n)_n, \lim_{n \rightarrow \infty} x_n = x_0 \Rightarrow \lim_{n \rightarrow \infty} f(x_n) = l$$

• to prove that the limit of a function doesn't exist in $N(x_0)$, it suffices to find two consequences $(x_n)_n, (y_n)_n$ that tend to x_0 but $\lim f(x_n) \neq \lim f(y_n)$

$\lim_{n \rightarrow \infty} \sin(x)$ doesn't exist

$x_n = 2n\pi, y_n = \frac{\pi}{2} + 2n\pi$

$\lim_{n \rightarrow \infty} x_n = \lim_{n \rightarrow \infty} y_n = +\infty$

$\lim_{n \rightarrow \infty} f(x_n) = 0 / \lim_{n \rightarrow \infty} f(y_n) = 1$

→ indeterminate forms of limit:

$\frac{0}{0}, 0 \times \infty, \frac{\infty}{\infty}, +\infty - \infty, \infty^0, 0^0, 1^\infty$

• indeterminate form 1^∞ :

f, g defined in $N(x_0), f(x) \neq 1$ in $N(x_0) (x \neq x_0), \lim_{x \rightarrow x_0} f(x) = 1$

$\lim_{x \rightarrow x_0} g(x) = \pm \infty$

if $\lim_{x \rightarrow x_0} (f(x) - 1) g(x) = \lambda \in \mathbb{R}$

we have:

$\lim_{x \rightarrow x_0} (f(x))^{g(x)} = e^\lambda$

$\lim_{n \rightarrow \infty} \left(\frac{n+1}{n-1} \right)^{2n+3}$ (simplify this first)

$f(x) = \frac{n+1}{n-1}, g(x) = 2n+3$

$\lim_{n \rightarrow \infty} (f(x) - 1) g(x) = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n-1} - 1 \right) (2n+3) = -4 (\lambda)$

$\Rightarrow \lim_{n \rightarrow \infty} \left(\frac{n+1}{n-1} \right)^{2n+3} = e^{-4} (e^\lambda)$

(3) - Continuity:

• at a point:

$\lim_{x \rightarrow x_0} f(x) = f(x_0)$

• $\forall \varepsilon > 0, \exists \alpha > 0, \forall x \in D_f$:

$|x - x_0| < \alpha \Rightarrow |f(x) - f(x_0)| < \varepsilon$

$\lim_{x \rightarrow x_0} f(x) = \lim_{x \rightarrow x_0} f(x) = f(x_0)$

→ f is continuous at x_0 .

• on an interval:

• f continuous on $]a; b[$, if it is continuous at every point of $]a; b[$

• f on $[a; b]$, if it is - on $]a; b[$ on the right of a and on the left of b .

• f, g are continuous at x_0 , then $\alpha f, |f|, f+g, f-g, fg (\alpha \in \mathbb{R})$ are continuous at x_0

if $g(x_0) \neq 0 \Rightarrow \frac{f}{g}$ is continuous at x_0 .

• Sequential continuity:

f is continuous at $x_0 \in \mathbb{R} \iff$ for all $(x_n)_n \subset D_f$ such that: $\lim_{n \rightarrow \infty} x_n = x_0$

we have: $\lim_{n \rightarrow \infty} f(x_n) = f(x_0)$

• to prove the opposite it suffices to find a sequence $(x_n)_n$ that converges to x_0 but $f(x_n)_n$ doesn't.

• Extension by continuity:

$x_0 \in \mathbb{R}, f$ is defined on $N(x_0), x_0 \notin D_f$

if $\lim_{x \rightarrow x_0} f(x) = l \in \mathbb{R} \Rightarrow f$ admits an

extension by continuity at x_0 :

$f(x) = \begin{cases} f(x) & \text{if } x \neq x_0 \\ l & \text{if } x = x_0 \end{cases}$

• intermediate value theorem:

• f is defined on $[a; b]$ + continuous on it

if $f(a) f(b) < 0 \Rightarrow \exists c \in]a; b[: f(c) = 0$

Generalization: $f([a; b])$ is an interval

(f1) $\Rightarrow f([a; b]) = [f(a), f(b)] = [\inf f(x), \sup f(x)]$

(f2) $\Rightarrow f([a; b]) = [f(b), f(a)] = [\inf f(x), \sup f(x)]$
 $a, b \in \mathbb{R} \cup \{-\infty; +\infty\}$

(f3) $\Rightarrow f(]a; b[) =] \lim_{x \rightarrow a} f(x), \lim_{x \rightarrow b} f(x) [$

(f4) $\Rightarrow f(]a; b[) =] \lim_{x \rightarrow b} f(x), \lim_{x \rightarrow a} f(x) [$

* I is an interval in $\mathbb{R}$

$f: I \rightarrow \mathbb{R}$

is a continuous function

strictly monotone

$\Rightarrow f$ is bijective

f^{-1} exists

f^{-1} is continuous and strictly monotone the same way as f .

• Uniform continuity:

* I is an interval on $\mathbb{R}$, $f: I \rightarrow \mathbb{R}$

f is uniformly continuous on I if:

$\forall \epsilon > 0, \exists \delta > 0, \forall (x, y) \in I \times I$:

$|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$

in the definition of continuity δ depends on x and ϵ , but here δ depends on ϵ only

• Heine's theorem: (see example)

$f: I \rightarrow \mathbb{R}$ (I is closed interval)

f is continuous on $I \Rightarrow f$ is uniformly continuous in I

(4) - Differentiability of functions:

$f: J \rightarrow \mathbb{R}, c \in J$

f is differentiable at c , if $\exists l \in \mathbb{R}$:

$$\lim_{n \rightarrow c} \frac{f(n) - f(c)}{n - c} = l$$

if it is useful to use $h = n - c$, it will become:

$$\lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h} = l$$

l is called the derivative of f at $n=c$ ($f'(c)$)

• f is differentiable at $n=c \Leftrightarrow f'(c) = f'(c)$

• Geometric interpretation:

• f is differentiable at c , the equation of the tangent at point $(c, f(c))$ is: $y = f'(c)(x - c) + f(c)$

• $\lim_{n \rightarrow c} \frac{f(n) - f(c)}{n - c} = 0 \Rightarrow$ the tangent is horizontal parallel to (Ox)

• $\lim_{n \rightarrow c} \frac{f(n) - f(c)}{n - c} = \infty \Rightarrow$ the tangent is vertical to (Oy)

• $f'(c) \neq f'_e(c) \Rightarrow (c, f(c))$ is an angular point

• f is differentiable at $a \Rightarrow f$ is continuous at a

• f is not continuous at $a \Rightarrow f$ is not differentiable at a

$$(g \circ f)'(x) = g'(f(x)) f'(x)$$

$$(f^n)'(n) = n f^{n-1} f'(n)$$

• Derivative of the inverse function:

$$(f^{-1}(y))' = \frac{1}{f'(f^{-1}(y))} \text{ if } f'(y) \neq 0$$

• $f''(c)$ is the second derivative of $f(c)$

• Derivative of order n ($n \geq 1$):

$$f^{(n)}(c) = f^{(n-1)}(c)$$

• f is of class $C^1 \Rightarrow \begin{cases} f \text{ is differentiable on } I \\ f' \text{ is continuous on } I \end{cases}$

• f is of class $C^n \Rightarrow \begin{cases} f \text{ is } n\text{-times diff... on } I \\ f' \text{ is continuous on } I \end{cases}$

• Leibniz formula:

• if f, g are n -times diff... on I , then fg is also n -times diff with

$$(fg)^{(n)}(x) = \sum_{k=0}^n C_n^k f^{(n-k)}(x) g^{(k)}(x)$$

(at point $(c) \rightarrow$ replace n by c)
($c \in I$)

• Mean value theorem:

→ Rolle's theorem:

$a, b \in \mathbb{R}, I = [a, b]$, f is continuous and differentiable on I , $f(a) = f(b)$, then:

$$\exists c \in]a, b[: f'(c) = 0$$

Generalization:

$a \in \mathbb{R}, I = [a; +\infty[$, continuous on I and differentiable on $]a; +\infty[$, $f(a) = \lim_{n \rightarrow +\infty} f(n)$

$$\exists c \in]a; +\infty[: f'(c) = 0$$

→ Mean Value theorem: MVT

Same conditions as (1) without $f(a) = f(b)$

$$\exists c \in]a, b[: f'(c) = \frac{f(b) - f(a)}{b - a}$$

Generalization:

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

→ L'Hopital's theorem

f, g continuous and diff at x_0 and at $N(x_0)$

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} \text{ is}$$

indetermined

$$\text{form } \frac{0}{0}, \frac{\infty}{\infty} \Rightarrow \lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = l$$

$$+ \lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = l$$

$$\bullet f'(x) = 0 \Leftrightarrow f = \text{cte}$$

$$\bullet f'(x) \geq 0 \Leftrightarrow f \nearrow$$

$$\bullet f'(x) \leq 0 \Leftrightarrow f \searrow$$

→ Taylor-Larange's theorem:

if f is a class of C^n on $[a, b]$ ($n \in \mathbb{N}^*$) and if $f^{(n+1)}$ exists on $[a, b]$

$$\exists c \in]a, b[:$$

$$f(b) = f(a) + \frac{b-a}{1!} f'(a) + \frac{(b-a)^2}{2!} f''(a) + \dots + \frac{(b-a)^n}{n!} f^n(a) + \frac{(b-a)(n+1)}{(n-n)!} f^{n+1}(c)$$

$$+ \dots + \frac{(b-a)^n}{n!} f^n(a) + \frac{(b-a)(n+1)}{(n-n)!} f^{n+1}(c)$$

• if $n=0$: MVT